

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

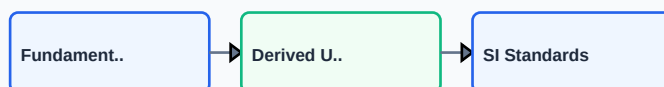
CHAPTER 1: MEASUREMENT

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Measurement

Learning Objectives:

- Understand the fundamental and derived quantities and their units.
- Know the rules to be followed while expressing physical quantities in SI units.
- Use Vernier caliper and screw gauge for small measurements.
- Find the weight of an object using a spring balance and know mass-weight differences.

TOPIC ROADMAP: MEASUREMENT STANDARDS

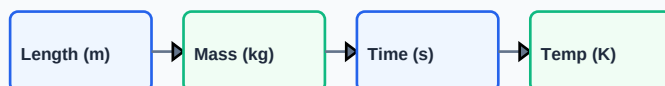
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Physical Quantity	A quantity that can be measured and in terms of which laws of physics can be described, consisting of a numerical value and a unit.
Fundamental Quantities	Quantities that cannot be expressed in terms of any other physical quantities, such as length, mass, time, temperature, electric current, luminous intensity, and amount of substance.
Derived Quantities	Quantities that can be expressed in terms of fundamental quantities, like area, volume, velocity, acceleration, force, and pressure.
Unit	The standard measure of a physical quantity used for comparison, such as metre, kilogram, or second.
Fundamental Units	The units of fundamental quantities, defined independently (e.g., metre for length, kilogram for mass, second for time).
Derived Units	The units used to measure derived quantities, expressed as combinations of fundamental units (e.g., m/s for velocity, kg/m ³ for density).
Metre	The SI base unit of length, defined as the distance travelled by light in vacuum during a time interval of $1/299,792,458$ of a second.
Kilogram	The SI base unit of mass, defined by taking the fixed numerical value of the Planck constant h to be $6.62607015 \times 10^{-34}$ when expressed in J s.
Second	The SI base unit of time, defined by taking the fixed numerical value of the caesium frequency $\Delta\nu_{\text{Cs}}$ to be 9,192,631,770 when expressed in Hz.
Kelvin	The SI base unit of thermodynamic temperature, defined by taking the fixed numerical value of the Boltzmann constant k to be 1.380649×10^{-23} when expressed in J K ⁻¹ .
Astronomical Unit	The mean distance between the center of the Earth and the center of the Sun, equal to 1.496×10^{11} metres.
Light Year	The distance travelled by light in vacuum in one year, equal to 9.46×10^{15} metres.
Parsec	Parallactic second, defined as the distance at which an arc of length 1 astronomical unit subtends an angle of one second of arc, equal to 3.08×10^{16} metres.
Least Count	The smallest value that can be measured accurately with a given measuring instrument (e.g., 0.1 mm for Vernier calipers, 0.01 mm for screw gauges).
Zero Error	The error in a measuring instrument when the zero mark of the measuring scale does not coincide with the reference zero index line under no-load conditions.
Pitch	The distance travelled by the spindle of a screw gauge per one complete rotation of the thimble / head scale.

Vernier Scale	A small, auxiliary graduated scale that slides along the main scale of an instrument, allowing readings to a fraction of a division.
Screw Gauge	A high-precision instrument used to measure dimensions in the millimetre range, such as wire diameter or metal sheet thickness, using a fine pitch screw.
Spring Balance	A device used to measure the weight of an object based on Hooke's Law, which states that the stretch of a spring is proportional to the force applied.

GLOSSARY: SI BASE UNITS STANDARD FLOW



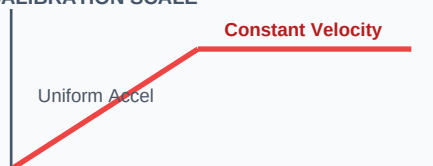
SI System of Units & Rules

The International System of Units (SI) is the modern metric system used globally in scientific studies.

Rules for Expressing SI Units:

- Units named after scientists start with a small letter (e.g., newton, kelvin, watt) but their symbols are capitalized (e.g., N, K, W).
- Symbols not named after scientists are in small letters (e.g., m, kg, s).
- No full stop or punctuation at the end of symbols (e.g., 50 m, not 50 m.).
- Symbols are never written in plural form (e.g., 10 kg, not 10 kgs).
- No degree sign for Kelvin temperature (e.g., 273 K, not 273 °K).
- Use negative powers instead of slash for compound units (e.g., ms^{-1} , not m/s).

VERNIER LEAST COUNT CALIBRATION SCALE



Scientific Notation & Measuring Instruments

In physics, we deal with extremely large values or extremely small values. We use scientific notation to write these efficiently.

Astronomical Distances:

- **1 Astronomical Unit (AU):** Mean distance from Earth to Sun = 1.496×10^{11} m.
- **1 Light Year (ly):** Distance travelled by light in vacuum in one year = 9.46×10^{15} m.
- **1 Parsec (pc):** Distance at which 1 AU subtends an angle of 1 arcsecond = 3.08×10^{16} m.

Instrument Characteristics:

- **Accuracy:** How close a measured value is to the true/accepted value.
- **Precision:** How close a series of measurements are to each other.
- **Least Count:** The smallest value that can be measured using an instrument.
- **Errors:** Systematic errors (instrumental calibration issues, personal bias) and Random errors (external variations, unpredictable factors).

WORKED: READING VERNIER CALIPERS DIMENSION



Vernier Caliper Measurement

A Vernier caliper is used to measure outer diameter, inner diameter, and depth of objects with high accuracy.

Least Count (LC) derivation:

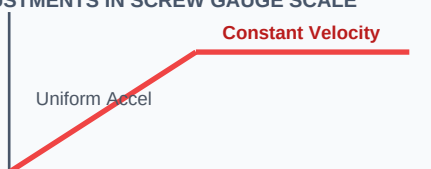
$LC = 1 \text{ Main Scale Division (MSD)} - 1 \text{ Vernier Scale Division (VSD)} = 1 \text{ mm} - 0.9 \text{ mm} = 0.1 \text{ mm} = 0.01 \text{ cm}.$

Zero Error and Zero Correction (ZC):

- **No Zero Error:** Zero of Vernier scale aligns exactly with zero of Main scale.
- **Positive Zero Error:** Vernier zero lies to the right of Main scale zero. ZC is negative. $(ZC = - (VC * LC))$.
- **Negative Zero Error:** Vernier zero lies to the left of Main scale zero. ZC is positive. $(ZC = + ((10 - VC) * LC))$.

Total Reading (TR) Formula: $TR = MSR + (VC * LC) \pm ZC$

HOTS: ZERO ERROR ADJUSTMENTS IN SCREW GAUGE SCALE



Screw Gauge Measurement

A screw gauge is used to measure dimensions in the range of 1/100th of a millimetre (0.01 mm), such as the thickness of a wire.

Pitch and Least Count (LC) Formula:

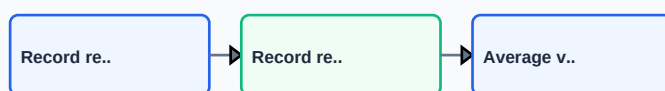
- **Pitch:** Distance moved by index line / Number of complete rotations = 1 mm.
- **Least Count:** Pitch / Number of Head Scale Divisions = 1 mm / 100 = 0.01 mm = 0.001 cm.

Zero Error and Correction (ZC):

- **Positive Zero Error:** Head scale zero lies below the index line. ZC is negative. ($ZC = - (HSC * LC)$).
- **Negative Zero Error:** Head scale zero lies above the index line. ZC is positive. ($ZC = + (100 - HSC) * LC$).

Total Reading (TR) Formula: $TR = PSR + (HSC * LC) \pm ZC$

PRACTICE: PHYSICAL BALANCE MEASUREMENT SHEET



Mass, Weight & Measuring Balances

Understanding the fundamental difference between mass and weight is vital in physics.

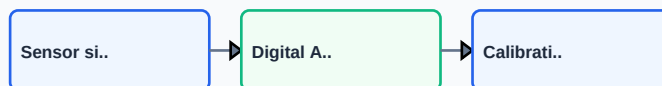
Mass vs Weight Differences:

- **Mass:** Quantity of matter contained in a body. Scalar quantity. Constant everywhere. Measured in kg. Measured using a physical balance.
- **Weight:** Gravitational force acting on the body's mass ($W = m.g$). Vector quantity. Varies with location. Measured in Newtons (N). Measured using a spring balance.

Types of Measuring Balances:

- **Common Beam Balance:** Compares sample mass with standard reference masses.
- **Physical Balance:** Used in labs for high precision (milligrams range).
- **Digital/Electronic Balance:** Fast, accurate, electronic sensor display.
- **Spring Balance:** Works on Hooke's Law (force extension relationship) to measure weight directly.

APPLICATION: INDUSTRIAL CALIBRATION IN ELECTRONICS



Worked Numerical Problems

Review these standard board examination worked examples to learn proper step layout structures:

Example 1: A Vernier caliper has a main scale reading of 4.5 cm and a Vernier coincidence of 6. If the positive zero error is 0.03 cm, calculate the corrected reading.

Step-by-step Solution:

Given: MSR = 4.5 cm VC = 6 LC = 0.01 cm Zero Error = +0.03 cm Zero Correction (ZC) = -0.03 cm Formula: TR = MSR + (VC * LC) + ZC Calculation: TR = 4.5 + (6 * 0.01) - 0.03 TR = 4.5 + 0.06 - 0.03 TR = 4.53 cm.

Example 2: A screw gauge has a pitch of 1 mm and 100 divisions on its head scale. The thimble shows a pitch scale reading of 3 mm and head scale coincidence of 45. Find the corrected reading if the instrument has a negative zero error of 0.04 mm.

Step-by-step Solution:

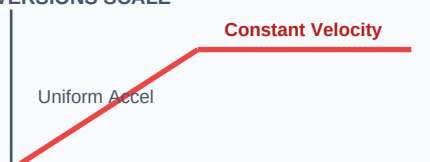
Given: Pitch = 1 mm Head scale divisions = 100 LC = Pitch / 100 = 0.01 mm PSR = 3 mm HSC = 45 Zero Error = -0.04 mm Zero Correction (ZC) = +0.04 mm Formula: TR = PSR + (HSC * LC) + ZC Calculation: TR = 3 mm + (45 * 0.01 mm) + 0.04 mm TR = 3 + 0.45 + 0.04 TR = 3.49 mm.

Example 3: Convert an astronomical distance of 2.5 light years into standard SI units (meters).

Step-by-step Solution:

Given: Distance = 2.5 light years (ly) 1 light year = 9.46×10^{15} m Formula: Distance in meters = Distance in ly * (9.46×10^{15} m/ly) Calculation: Distance = $2.5 * 9.46 \times 10^{15}$ Distance = 23.65×10^{15} m = 2.365×10^{16} m.

UNIT PREFIXES AND CONVERSIONS SCALE



HOTS & Practice Worksheet

HOTS Analytical Questions:

Q1: Why are astronomical distances expressed in light years or parsecs instead of standard SI units like meters?

Strategy/Answer: Astronomical distances (e.g. between stars and galaxies) are incredibly large. Writing them in meters yields extremely large numbers (e.g. Earth to Sun is 149,600,000,000 meters) which are prone to clerical and calculation errors. Light years (distance traveled by light in a year) and parsecs simplify these numbers to manageable quantities.

Q2: A spring balance calibrated on Earth is taken to the Moon to measure the mass and weight of a rock. Explain how the readings of the spring balance will change.

Strategy/Answer: A spring balance measures weight (gravitational force, $W = m \cdot g$) rather than mass. Since acceleration due to gravity on the Moon ($g_{\text{moon}} \approx 1.63 \text{ m/s}^2$) is 1/6th of that on Earth ($g_{\text{earth}} \approx 9.8 \text{ m/s}^2$), the weight reading on the spring balance will decrease to 1/6th of its Earth value. The rock's actual mass remains unchanged, but a spring balance cannot measure mass directly without recalibration.

Independent Practice Worksheet:

Q1: Differentiate between fundamental and derived physical quantities, giving three examples of each.

Q2: State the rules for writing SI units and their symbols (capitalization, plurals, punctuation) as specified by BIPM.

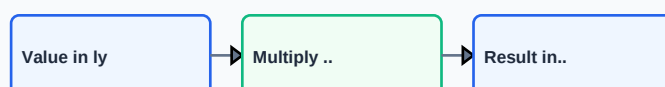
Q3: A screw gauge has a PSR of 3 mm and HSC of 42. If it has a negative zero error of 5 divisions, find the thickness of the wire.

Q4: What is the difference between physical balance and digital balance in terms of sensitivity and accuracy?

Q5: Why is a degree sign omitted when writing temperatures in Kelvin (e.g., 273 K, not 273 °K)?

Q6: A Vernier caliper has 10 divisions on the Vernier scale which coincide with 9 divisions on the main scale (1 division = 1 mm). If the jaws are closed and the Vernier zero is to the left of the main scale zero by 2 divisions, calculate the zero error and zero correction.

PRACTICE: UNIT CONVERSIONS FLOW



Revision Summary & Strategy

Real-World Applications:

Application Case: Scientific Laboratories

Vernier calipers and screw gauges are standard instruments to measure the diameters of spherical lenses, capillaries, and wires with high precision.

Application Case: Aviation and Space Navigation

Astronomical units and light years are used to coordinate orbits, rocket vectors, and deep space telemetry without rounding errors.

Application Case: Metallurgical and Wire Manufacturing Industry

Used to gauge wire thicknesses, sheet metal profiles, and precision components in machining and aerospace fabrication to ensure tolerances.

Syllabus Checklist:

- ☐ Memorized base SI quantities, units, and writing rules (Page 3).
- ☐ Reviewed Vernier caliper least count & errors (Page 5).
- ☐ Reviewed Screw gauge least count & errors (Page 6).
- ☐ Solved worked examples & worksheet problems (Page 8 & 9).

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: MEASUREMENT CONCEPTS CYCLE

